



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PASCHIMANCHAL CAMPUS**

**A Final Report
On
Smart AI-Based Water Quality Monitoring System
Using
IoT and Machine Learning**

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ABSTRACT

This project presents the design and implementation of Smart AI-Based Water Quality Monitoring System Using IoT and Machine Learning to monitor and assess drinking water quality in real time. The hardware layer utilizes an ESP32 microcontroller interfaced with pH and Total Dissolved Solids (TDS) sensors that send continuous readings to a cloud-based (back-end) service. To ensure high-reliability safety grading, it uses a dual-layered pipeline that includes a Random Forest classifier for predicting safety levels statistically, and the WHO rules engine to override the predictions based on deterministically safe/unsafe values.

The model was trained using a hybrid dataset of 4,451 samples that combines both global and local field data. Results demonstrate high performance with a test accuracy level of 99.78% across four safety grades: GOOD, MODERATE, POOR, and CRITICAL. The system includes a React dashboard that provides live analytics and critical hazard alerts. Ultimately, the system provides a low-cost, scalable, and easily interpretable solution suitable for schools, hostels, and small communities in Nepal where laboratory testing is often inaccessible.

Keywords: AI/ML, TDS sensor, pH sensor, ESP-32, MERN, Random Forest, Logistic Regression

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
TDS	Total Dissolved Solids
IoT	Internet of Things
ESP	Espressif Systems Platform
MERN	MongoDB, Express.js, React, and Node.js
WQI	Water Quality Index
Wi-Fi	Wireless Fidelity
ADC	Analog Digital Converter
WHO	World Health Organization
ppm	Parts Per Million
pH	Potential of Hydrogen
JSON	JavaScript Object Notation
ODM	Object Data Modeling
HTTP	HyperText Transfer Protocol
DOM	Document Object Model
IDE	Integrated Development Environment
API	Application Programming Interface
CSV	Comma-Separated Values
HGB	Histogram Gradient Boosting
BIS	Bureau of Indian Standards

1. INTRODUCTION

1.1. Background

Water is a fundamental necessity for sustaining life, and access to clean and safe drinking water is critical for maintaining public health. However, in many parts of the world (especially in developing nations like Nepal), the availability of real-time and affordable water quality monitoring solutions remains limited. Water sources in rural areas, hostels, and even some urban communities are often unregulated or tested manually at irregular intervals, making it difficult to detect contamination before it affects human health.

Traditional methods of water quality testing rely heavily on manual sampling followed by laboratory-based analysis. These processes are not only time-consuming and expensive, but they also require specialized knowledge and equipment, limiting their practical application in remote or resource-constrained settings. In many cases, water quality testing devices like TDS meters and pH testers merely display numeric values without context, requiring interpretation by trained individuals.

With the advancement of low-cost IoT sensors and the increasing power of Artificial Intelligence (AI), particularly in data classification and prediction, there now exists a promising opportunity to automate and simplify the monitoring of water quality. By integrating sensor data with intelligent algorithms, we can create real-time systems that analyze water parameters and inform users about safety levels instantly, accurately, and without requiring technical expertise.

1.2. Motivation

During various observations and informal surveys conducted in school hostels, rural communities, and public tap systems, a recurring problem came to light: users lacked both awareness and tools to determine whether the water they used daily was safe for consumption. Even when sensors were available, such as digital TDS or pH meters, the values they produced were either misunderstood or ignored due to a lack of context and technical background. In some instances, this led to the prolonged consumption of low-quality water, resulting in waterborne illnesses and other health issues.

This motivated us to design an AI-based water quality monitoring system that not only collects real-time sensor data but also interprets it through machine learning models to deter-

mine water safety. The system is capable of classifying water as “Safe” or “Unsafe” based on multi-parameter analysis and will notify users through a simple, user-friendly interface or alert mechanism. The aim is to eliminate the guesswork from water testing and empower users, from students in hostels to households in remote areas, with timely, accurate information about their water quality.

By making the system affordable, portable, and intelligent, we intend to contribute toward better water safety, reduced health risks, and increased environmental awareness. Moreover, this project serves as an example of how interdisciplinary engineering, combining hardware, software, and AI can be used to tackle critical societal issues in practical and impactful ways.

1.3. Problem Statement

The currently available water quality monitoring solutions are often expensive, time consuming, and difficult for common users to understand. Traditional methods such as manual sampling and laboratory testing can delay the detection of contaminated water, while devices such as TDS and pH sensors only provide raw readings that may be confusing for non-technical users. In many cases, there is also a lack of real-time alerts to warn users when water becomes unsafe for consumption. Additionally, affordable and intelligent water quality monitoring systems are still not easily accessible in schools, hostels, rural communities, and public areas.

1.4. Objectives

- To train and optimize a machine learning model using sensor data (pH and TDS) to classify water quality as Good, Moderate, Poor, and Critical.
- To integrate the trained model with an ESP32 microcontroller for real-time prediction and alert generation through LEDs or buzzers.

2. LITERATURE REVIEW

The study presents an intelligent IoT-based system for monitoring drinking water quality for tourists in Bangladesh. Using an ESP32 microcontroller and sensors for pH, turbidity, TDS, and temperature, real-time data were collected and analyzed from 3,178 samples. Machine learning models were applied, with an artificial neural network achieving 92.66 percent of accuracy. Results revealed widespread contamination, demonstrating the system's effectiveness as a scalable alternative to conventional testing methods.[1]

This paper presents a review of IoT-based water quality monitoring systems using wireless sensor networks and LPWAN/Zigbee communication protocols. These systems monitor key parameters such as pH, turbidity, and dissolved oxygen, and increasingly integrate machine learning for anomaly detection and prediction. However, challenges such as sensor calibration drift, heterogeneous data integration, and deployment costs remain significant in low-resource settings. The study concludes that such systems are essential for sustainable water management and supporting global environmental goals, including SDGs.[2]

According to Abbas et al. (2024), machine learning models can reliably predict WQI from limited input data, often with high accuracy, eliminating the need for full lab-based testing or large number of physical sensors. This supports our decision to use only simple sensors (pH + TDS) and ML in our fish farm monitoring system: we plan to compute a water quality index in real-time via ML, offering a low-cost yet effective monitoring solution. The paper also shows that even with incomplete data, ML outputs stay robust which increases confidence in using ML-based WQI in our local context.[3]

This paper gives a reference for combining IoT + ML + soft sensors to monitor water quality beyond simple parameters, exactly like my idea to estimate more water-quality indicators without expensive hardware. It shows that even when direct sensors are lacking for certain metrics (like microbial contamination), ML + IoT can provide useful predictions.[4]

3. RELATED THEORY

3.1. Water Quality Parameters

3.1.1. Potential of Hydrogen (pH)

The pH value measures the activity of hydrogen ions in a solution and indicates whether the water is acidic, neutral, or alkaline. The pH scale spans from 0 to 14, with 7 representing neutrality. Drinking water with a pH below 5.5 is considered dangerously acidic and may corrode pipes, while a pH above 9.5 is considered alkaline and can cause scaling, bitter taste, and gastric discomfort. The WHO recommends a pH range of 6.5 to 8.5 for safe drinking water.

3.1.2. Total Dissolved Solids (TDS)

TDS denotes the total concentration (in mg/L or ppm) of dissolved inorganic salts and small amounts of organic matter present in water. It is a strong indicator of mineral content, hardness, and general drinkability. According to WHO and BIS IS:10500 guidelines [2], drinking water with TDS less than 300 ppm is considered excellent, between 300–600 ppm is acceptable, between 600–900 ppm is permissible only when no better source is available, and above 900 ppm is considered unfit for human consumption.

3.2. Hardware Components

3.2.1. ESP32 Microcontroller

The ESP32 is a powerful, low-cost microcontroller with integrated Wi-Fi and Bluetooth capabilities, developed by Espressif Systems. It is widely used in IoT (Internet of Things), home automation, wireless sensing, and embedded applications due to its versatility, dual-core performance, and low power consumption. The ESP32 operates based on a dual-core CPU architecture. One core typically handles user application logic while the other manages wireless communication (Wi-Fi/Bluetooth), allowing for efficient multitasking. The device can be programmed using the Arduino IDE, MicroPython, or ESP-IDF (Espressif IoT Development Framework).

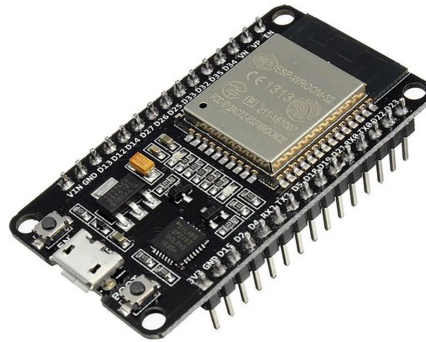


Figure 3.1: ESP32

3.2.2. Gravity Analog pH Sensor

A pH sensor is a device that is used for determining the acidity or alkalinity of a solution by measuring its hydrogen ion concentration, i.e, pH value, which ranges from 0-14. The pH value below 7 indicates that the solution is acidic, value of 7 indicates neutral, and above 7 indicates that the solution is basic or alkaline. The onboard voltage regulator chip supports the wide voltage supply range of 3.3V to 5.5V, which is compatible with the 5V and 3.3V main control board. The most common type of pH sensor consists of a glass electrode that responds to H activity and a reference electrode that provides a stable voltage value for comparison. The sensor measures the difference between these electrodes and converts the output voltage to pH value using the Nernst equation using its inner electronics, which comprises an op-amp, ADC, a small microcontroller, etc. These components work together to convert the small voltage signals sensed from electrodes into a stable pH value.

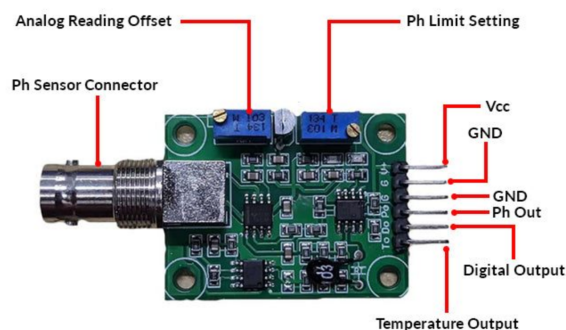


Figure 3.2: pH sensor

3.2.3. Gravity Analog TDS Sensor

A TDS (Total Dissolved Solids) sensor is a device that is used to measure the concentration of dissolved combined contents of all inorganic and organic substances present in water like salts, minerals, etc. The TDS is measured in parts per million(ppm). The higher the TDS value, the more soluble solids are dissolved in the water, and the less clean the water is. According to the WHO, a TDS value below 500 ppm is considered safe for drinking purposes, and greater than 1000 ppm is not suitable for human consumption. It supports 3.3-5.5V voltage input and 0- 2.3V analog voltage output, which makes it compatible with 5V or 3.3V control systems or boards. It is used widely for checking drinking water quality and areas such as agriculture, industries, food and beverages and so on.

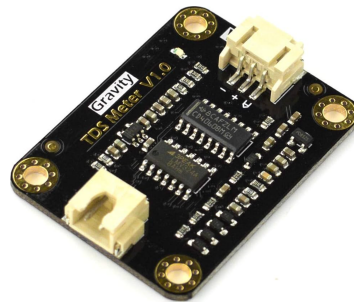


Figure 3.3: TDS sensor

3.3. Software Components

3.3.1. Database

- **MongoDb**

MongoDB is a NoSQL, document-oriented database designed for flexibility, scalability, and performance. Unlike traditional relational databases, MongoDB stores data in JSON-like documents, allowing for a more natural mapping to application objects and real-time sensor data. Its schema-less nature allows us to adapt quickly to changes in data structure without complex migrations. Paired with Mongoose, an ODM (Object Data Modeling) library for Node.js, MongoDB offers powerful querying, validation, and model-based interaction, simplifying backend development. MongoDB's scalability and ability to handle large volumes of incoming sensor data make it a reliable choice for our project, ensuring consistent performance as the system grows and evolves.

- **Backend Node and Express.js**

Node.js, as a JavaScript runtime, enables rapid and scalable development on the server side. It uses the V8 engine found in Chrome and makes non-blocking operations as well as event-driven systems possible. This is perfect for real-time applications, like our Water Quality Monitoring System. Express.js is a minimalist web framework that works on Node.js. It is simple and unadorned and covers the structure of routing, HTTP requests, middleware, and even the setup of the server itself. With Express, we will be able to receive and store sensor data, run AI models to predict Water Quality Index(WQI), and push classification results to the frontend in real-time. Together, both Node.js and Express.js serve as a steady backbone for our project. With these technologies, we are sure that data will flow smoothly, get api calls in no time, and have real-time interactions with AI modules, hardware devices, and the frontend.

3.3.2. Frontend React js

React, developed by Facebook, is a free and open-source JavaScript library for building dynamic user interfaces. React's component-based architecture aligns well with our goal of creating reusable UI components for seamless interaction. With its declarative syntax and optimization through a virtual DOM, React will enhance the performance of our system, ensuring smooth updates in response to data changes. Additionally, the features like Hooks and React Router manage the state and side effects and efficient client-side routing, respectively, which further improve the overall functionality and the code organization of the app.

3.3.3. Platform io

Platform IO is an open-source ecosystem designed for embedded development, facilitating cross-platform programming for microcontrollers and embedded systems. It offers a unified development experience, supporting various hardware platforms, 8 development boards, and frameworks. Platform IO simplifies tasks such as configuring, building, and deploying firmware, enhancing efficiency for developers working on embedded projects. With its powerful command-line interface and IDE integration, Platform IO provides a seamless environment for coding, debugging, and managing projects.

3.4. Machine Learning Algorithms

3.4.1. Random Forest Classifier

Random Forests train many decision trees on different random samples and feature subsets, then take a majority vote. This makes them much more accurate and stable than a single tree because individual tree errors cancel out.

They fit this project well because the WHO rules for pH and TDS are simple, axis-aligned thresholds that decision trees represent naturally. A forest of trees reproduces those boundaries reliably while smoothing out sensor noise. Random Forests also give feature-importance scores (how much pH and TDS each matter), so you get some built-in explanation of decisions.

Training 300 trees on 3,115 samples takes about 5–10 seconds on a laptop, and each prediction is under 1 ms, which easily meets the ESP32’s 6-second measurement cycle.

3.4.2. Logistic Regression (Baseline)

Logistic Regression learns a linear decision boundary in feature space. It was included in the model selection comparison specifically because it is expected to fail on this problem: the WHO decision regions are rectangular boxes in the pH-TDS plane, and no linear classifier can carve out a rectangle from its surrounding classes. Its poor validation performance (50.34% accuracy) confirms that the classification problem is genuinely non-linear, which in turn justifies the choice of a tree-based ensemble.

3.4.3. Histogram Gradient Boosting (Baseline)

Histogram Gradient Boosting (HGB) is a modern sequential boosting algorithm that constructs trees one at a time, where each new tree is trained to correct the residual prediction errors of all previous trees [5]. It is state-of-the-art on many tabular datasets and achieved a 97.98% validation F1, which in comparison is a strong result, but still approximately 1.8 percentage points below Random Forest.

HGB is mentioned only as a comparison baseline. It was **not** selected for deployment. The performance gap arises because the WHO boundaries are axis-aligned thresholds that a parallel bagging ensemble (RF) recovers more directly than a sequential correction chain.

3.5. Standardisation: WHO and BIS IS:10500 Guidelines

The classification of water quality in this project is based on internationally accepted drinking water standards provided by WHO and BIS. These standards help determine whether water is safe or unsafe for human consumption using pH and TDS values.

According to the WHO drinking water guidelines, the recommended pH range for safe drinking water is 6.5 to 8.5. Water with a pH lower than 6.5 is acidic and may damage pipes or affect health, while water with a pH higher than 8.5 may contain excess minerals and create scaling problems.

For Total Dissolved Solids (TDS), the BIS IS:10500 standard is used. Based on this standard, water below 300 ppm is considered excellent for drinking. Water between 300 and 600 ppm is acceptable, while 600 to 900 ppm is allowed only in special situations where better water sources are not available. TDS values above 900 ppm are treated as unsafe for drinking purposes.

4. SYSTEM ARCHITECTURE

4.1. System Block Diagram

The complete block diagram is illustrated in Figure 4.1. The Edge Layer (ESP32) receives the analog inputs, the Backend Layer (Node.js + MongoDB) stores and manages the data, the AI Layer (FastAPI + scikit-learn) processes the classification on the basis of the data, and the Frontend Layer (React) displays the data in a real-time dashboard.

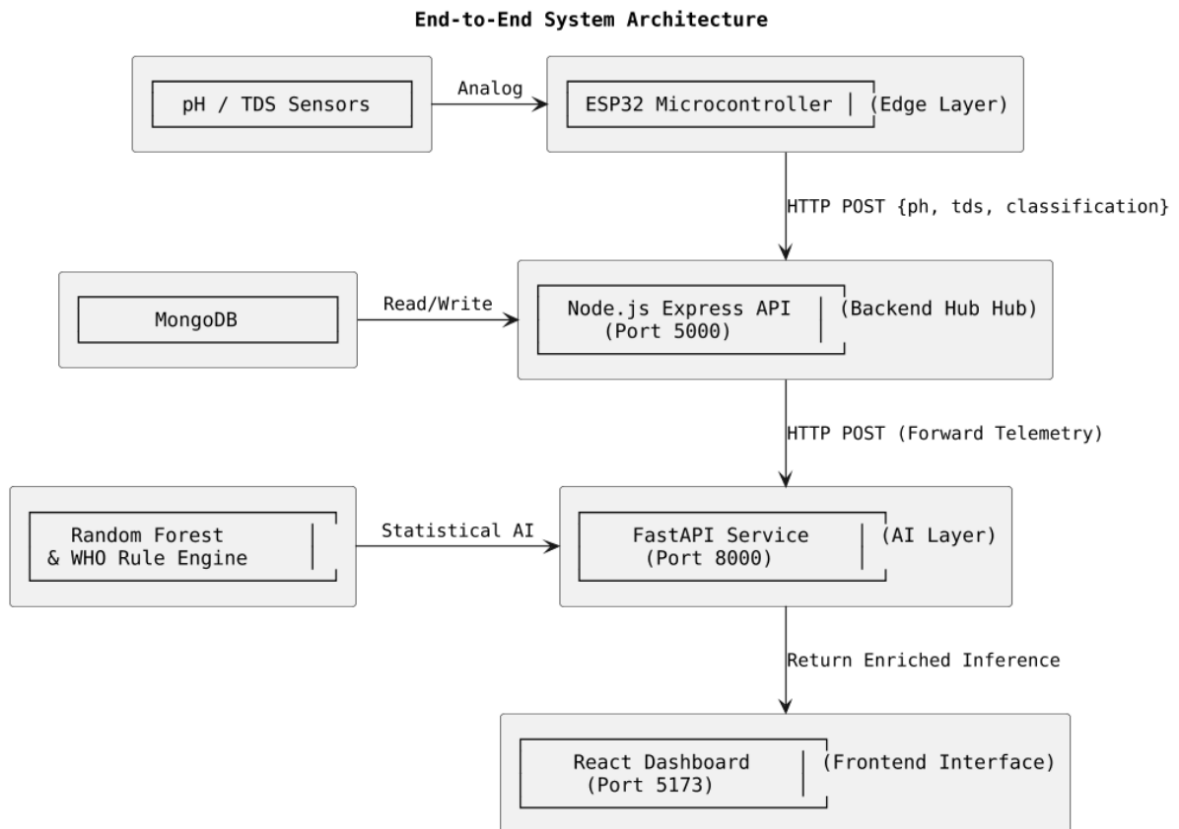


Figure 4.1: System Block Diagram

4.2. Activity Diagram

The activity flow diagram shown in Figure 4.2 shows what happens to data during a single cycle.

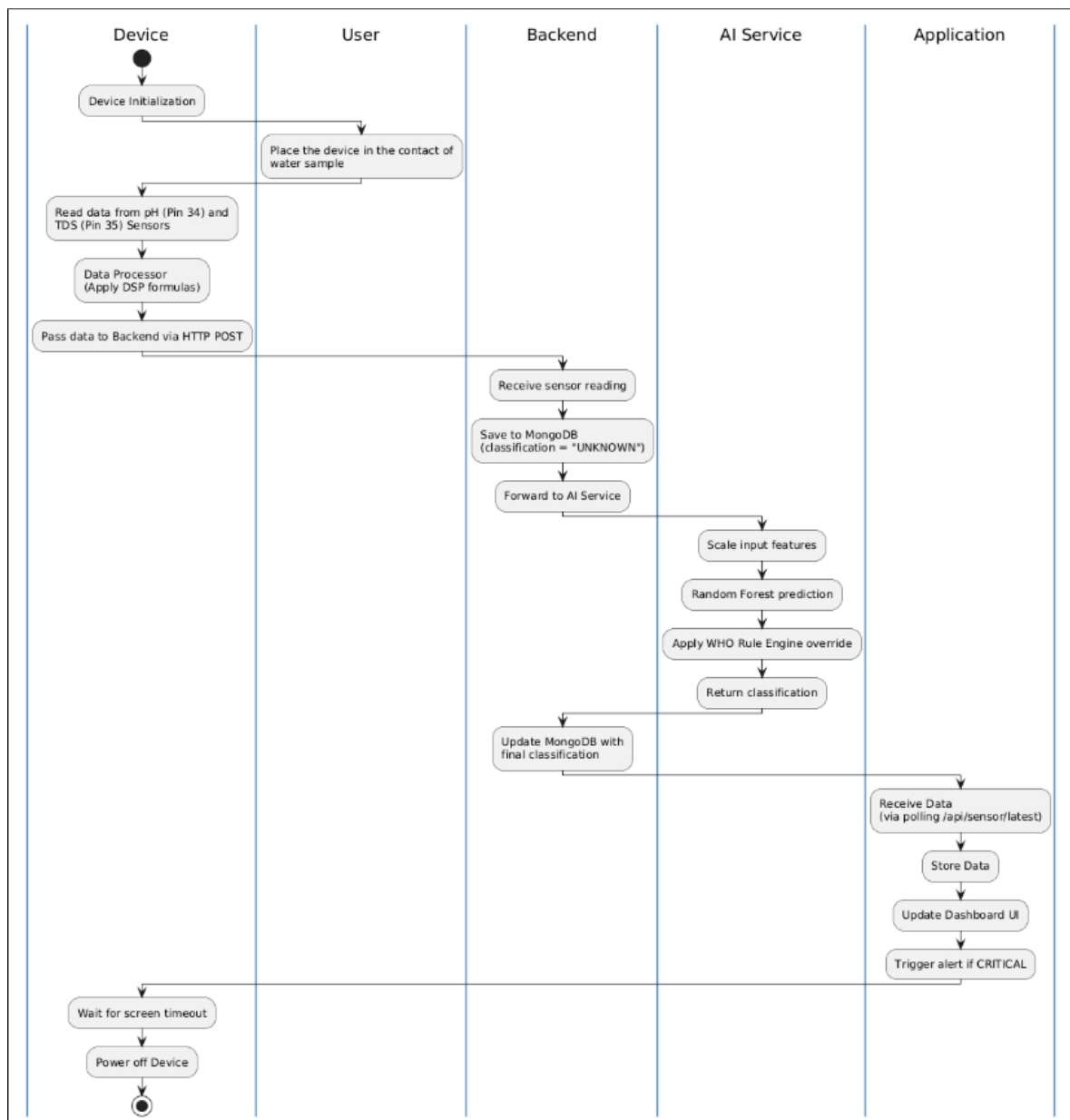


Figure 4.2: Activity Diagram

5. Methodology

This chapter describes the systematic approach used to build the Smart AI-based Water Quality Monitoring System Using IoT and Machine Learning, covering the integration of IoT devices and the machine learning process in the full-stack MERN technology stack.

5.1. Workflow Description)

The system operates in a continuous "Sense-Predict-Act" cycle designed for high-frequency monitoring. When probes are submerged, the ESP32 captures analog telemetry, converts it into physical units, and transmits it via HTTP POST to a Node.js backend hub. The backend acts as an orchestrator, concurrently saving data to MongoDB and forwarding it to a FastAPI intelligence layer for classification. The result is then broadcast to a React dashboard, which provides real-time visualization and triggers localized alarms if hazards are detected. This end-to-end workflow ensures that raw sensor data is translated into an actionable safety status within seconds.

5.2. Data Collection

To ensure a robust predictive model capable of handling both theoretical patterns and hardware-specific noise, a hybrid dataset of 4,451 unique samples was utilized.

- **Kaggle Public Dataset:** 3,276 rows were sourced from a foundational water quality dataset. A critical correction was applied to the Solids column (divided by 100) to align the data with realistic ppm units consistent with WHO guidelines.
- **ESP32 Sensor Dataset:** 1,676 readings were collected directly using the prototype hardware. The samples for this additional set were collected using tap water, distilled water, spring water, and contaminated water(with diluted lemon solution, soap solution, and salt) in order to cover the pH readings from 1.66 to 13.3 and the TDS value range ranging from 2.7 to 2575 ppm.
- **Ground-truth Protocol:** All samples were labeled using the WHO 2022 Guidelines, ensuring that the relationship between analog voltage and physical safety tiers remained consistent across both data sources.

5.3. Data Logging Format

Each data entry was recorded in a structured format to ensure consistency during preprocessing and real-time inference:

- **pH:** Numeric value (0–14) captured via the analog signal unit.
- **TDS:** Numeric value in ppm, derived from voltage via a cubic polynomial formula.
- **Source:** Tagged as Kaggle or Sensor for traceability.
- **Classification:** The assigned safety tier (**GOOD**, **MODERATE**, **POOR**, **CRITICAL**).

5.4. Data Preprocessing

The collected data underwent a multi-stage preprocessing phase to enhance model robustness:

- **Null Handling and Cleaning:** Rows with missing pH values were dropped rather than imputed to avoid introducing false "GOOD" readings into a safety-critical system.
- **Feature Scaling:** We applied StandardScaler (Z-score normalization) to the pH and TDS features. This was mandatory because pH (0–14) and TDS (0–2500+) operate on different numeric scales without scaling; the model would unfairly prioritize the larger TDS values.
- **WHO Hybrid Rule:** To prioritize public health, we implemented a Safety-First Priority Logic. The final label for any sample is the maximum (worst) of the pH and TDS classes:

$$\text{Final_Label} = \max(\text{pH_Class}, \text{TDS_Class})$$

This ensures that if water is chemically dangerous (e.g., pH = 4.0) but physically clear (low TDS), it is correctly flagged as **CRITICAL**.

TDS range (ppm)	Class	Code
TDS < 300	GOOD	0
$300 \leq \text{TDS} < 600$	MODERATE	1
$600 \leq \text{TDS} < 900$	POOR	2
TDS ≥ 900	CRITICAL	3

Figure 5.1: TDS classification thresholds(WHO/BIS IS:10500)

pH range	Class	Code
$6.5 \leq \text{pH} \leq 8.5$	GOOD	0
$6.0 \leq \text{pH} < 6.5$ or $8.6 \leq \text{pH} \leq 9.0$	MODERATE	1
$5.5 \leq \text{pH} < 6.0$ or $9.1 \leq \text{pH} \leq 9.5$	POOR	2
$\text{pH} < 5.5$ or $\text{pH} > 9.5$	CRITICAL	3

Figure 5.2: pH classification thresholds (WHO Guidelines 2022)

5.5. Model Generation

To develop a high-accuracy classification engine, three modeling approaches were evaluated on a 10% validation set:

- **Baseline (Logistic Regression):** Achieved only 50.34% accuracy, failing to capture the non-linear boundaries of water safety standards.
- **Random Forest Classifier:** Selected as the production model after achieving a validation accuracy of 99.78% and a weighted F1-score of 0.9977. This ensemble of 300 decision trees proved most effective at mapping axis-aligned threshold rules For example, $\text{pH} < 5.5 \rightarrow \text{CRITICAL}$.
- **Deployment:** The final model and scaler were serialized using joblib into .pkl files and hosted via **FastAPI** to provide low-latency inference endpoints.

5.6. Result and Observation

The performance of the classification system was evaluated using standard metrics, focusing on precision and recall for hazard detection: $\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100\%$

Observations confirmed that Random Forest significantly outperformed linear models by successfully representing the "box-shaped" safety boundaries defined by WHO standards. The system demonstrated Noise Robustness, maintaining high accuracy even during simulated sensor drift, provided the hardware undergoes periodic calibration with standard buffer solutions.

5.7. Web Application Development

The primary interface was developed as a React web dashboard, providing a centralized view of the water supply status.

- **Simplified Interpretation:** The dashboard is the visual representation of real-time pH and TDS values in a format easily understandable for non-technical individuals at a single glance. The dashboard represents the actual pH and TDS indicators visually in the form of intuitive, color-based indicators.
- **Safety Logic:** The app employs a double layer of safety measures that ensure automatic refreshing of the backend for updates every 5 seconds. If it returns CRITICAL status, the app immediately overrides normal functionality by triggering both acoustic alerts(buzzer) from the computer and notifications from the browser.

6. REQUIREMENT

This section details the key technical, software, and hardware components required for the effective development, deployment, and maintenance of the proposed system.

6.1. Software and Development Environment

The system utilizes a modern full-stack architecture to manage the flow of sensor data and AI predictions.

6.1.1. Development Tools:

- **PlatformIO:** Used for embedded C++ development and managing ESP32 libraries .
- **VS Code:** Primary IDE for backend, frontend, and AI service development.
- **Postman:** Employed for testing RESTful API endpoints.
- **MongoDB Compass:** A GUI-based tool used for real-time visualization and management of the telemetry stored in the NoSQL database.
- **Version Control:**Git with GitHub or GitLab for source code management, collaboration, and version tracking.

6.1.2. Programming Languages:

- **C++ (Arduino framework):** Used for programming the ESP32 firmware logic.
- **JavaScript (ES6+):** Utilized for both the Node.js backend and React frontend.
- **Python:** Used for machine learning model training and the FastAPI service.

6.2. Libraries and Frameworks

6.2.1. MERN Stack:

- **MongoDB:** NoSQL database for storing telemetry and classification history.

- **Express.js and Node.js:** Manages the backend hub and hardware synchronization.
- **React and Vite:** Provides the dynamic dashboard and visualization components.

6.2.2. Intelligence Layer:

- **FastAPI and Uvicorn:** Serves the Random Forest model with high performance and data validation.
- **Scikit-learn:** The primary machine learning library used for implementing the Random Forest Classifier, StandardScaler, and performance evaluation metrics.
- **Pandas and NumPy:** Essential libraries for data manipulation, cleaning the Kaggle dataset, and performing vectorized mathematical operations on sensor telemetry.
- **Joblib:** Facilitates the serialization and loading of .pkl model and scaler files.

6.2.3. Hardware Libraries:

- **WiFi.h and HTTPClient.h:** Enable ESP32 to connect to local networks and send POST requests.
- **ArduinoJson:** A critical library for the ESP32 used to serialize sensor readings into a structured format compatible with the backend API.

7. EPILOGUE

7.1. Analysis of Source Output

- **RO/Distilled water:** As shown by the telemetry logs, the testing of a purified baseline stream showed the baseline sensitivity capability of the system. The sensors captured the perfect neutrality of the stream with a pH of 7.05 and a TDS reading of 0 ppm, signifying no presence of any conductive ions. The classification algorithm was able to map the pure state as a “GOOD” tier.

_id	ObjectId	ph Mixed	tds Mixed	classification String
2337	ObjectId('6a05923613e86cc...	7.01	0	"GOOD"
2338	ObjectId('6a05923d13e86cc...	7.24	0	"GOOD"
2339	ObjectId('6a05924413e86cc...	7.08	0	"GOOD"
2340	ObjectId('6a05924c13e86cc...	7.01	0	"GOOD"
2341	ObjectId('6a05925313e86cc...	7	0	"GOOD"
2342	ObjectId('6a05925b13e86cc...	7.06	0	"GOOD"
2343	ObjectId('6a05926213e86cc...	6.87	0	"GOOD"
2344	ObjectId('6a05926913e86cc...	7.03	0	"GOOD"
2345	ObjectId('6a05927113e86cc...	6.96	0	"GOOD"
2346	ObjectId('6a05927813e86cc...	7.06	0	"GOOD"

Figure 7.1: RO Water Data

- **Municipal Tap Water:** During the testing of municipal tap water, it was found that there was remarkable stability in ionic characteristics. The telemetry logs show a neutral pH fluctuating safely between 6.59 and 7.30, along with a moderate mineral footprint fluctuating between 180.71 ppm and 216.45 ppm. Since both parameters were firmly within the regulatory range of the WHO framework, the classification algorithm continuously classified it as GOOD.

	_id ObjectId	ph Mixed	tds Mixed	classification String
2362	ObjectId('6a0592e913e86cc...	7.3	180.71	"GOOD"
2363	ObjectId('6a0592ef13e86cc...	6.84	187.19	"GOOD"
2364	ObjectId('6a0592f613e86cc...	6.83	204.36	"GOOD"
2365	ObjectId('6a0592fd13e86cc...	7.05	210.1	"GOOD"
2366	ObjectId('6a05930513e86cc...	6.86	200.56	"GOOD"
2367	ObjectId('6a05930c13e86cc...	6.79	216.36	"GOOD"
2368	ObjectId('6a05931313e86cc...	6.63	215.5	"GOOD"
2369	ObjectId('6a05931913e86cc...	6.81	216.07	"GOOD"
2370	ObjectId('6a05932713e86cc...	6.74	216.45	"GOOD"
2371	ObjectId('6a05932d13e86cc...	6.59	215.66	"GOOD"
2372	ObjectId('6a05933313e86cc...	6.77	214.44	"GOOD"
2373	ObjectId('6a05933a13e86cc...	6.71	212.59	"GOOD"
2374	ObjectId('6a05934113e86cc...	6.64	212.25	"GOOD"

Figure 7.2: Municipal Tap Water Data

- **Natural Spring Water :** The testing of natural spring water pointed out that the system is capable of detecting changes in the parameters. Even though the TDS was very low and clean in terms of an average of 153.20 ppm, the natural fluctuations in the runoff led to a decrease in pH value between 6.31 and 6.49 which is below the recommended level for drinking purposes. Since the “Safety-First” priority maximum rule is embedded in the system, it correctly skipped the “GOOD” category and mapped as “MODERATE”.

	_id ObjectId	ph Mixed	tds Mixed	classification String
2316	ObjectId('6a0591aa13e86cc...	6.48	154.56	"MODERATE"
2317	ObjectId('6a0591b113e86cc...	6.49	154.27	"MODERATE"
2318	ObjectId('6a0591b813e86cc...	6.4	154.6	"MODERATE"
2319	ObjectId('6a0591bf13e86cc...	6.43	154.81	"MODERATE"
2320	ObjectId('6a0591c613e86cc...	6.41	152.88	"MODERATE"
2321	ObjectId('6a0591cc13e86cc...	6.4	153.24	"MODERATE"
2322	ObjectId('6a0591d313e86cc...	6.38	152.49	"MODERATE"
2323	ObjectId('6a0591db13e86cc...	6.38	152.73	"MODERATE"
2324	ObjectId('6a0591e213e86cc...	6.34	152.65	"MODERATE"
2325	ObjectId('6a0591e813e86cc...	6.34	152.52	"MODERATE"
2326	ObjectId('6a0591ee13e86cc...	6.33	152.53	"MODERATE"
2327	ObjectId('6a0591f413e86cc...	6.34	152.06	"MODERATE"
2328	ObjectId('6a0591fa13e86cc...	6.32	153.01	"MODERATE"
2329	ObjectId('6a05920013e86cc...	6.31	152.27	"MODERATE"
2330	ObjectId('6a05920713e86cc...	6.33	152.09	"MODERATE"

Figure 7.3: Spring Water Data

7.2. Limitations

The system, as currently deployed, carries five acknowledged limitations worth stating honestly. First, it measures only pH and TDS, meaning critical parameters such as turbidity, temperature, dissolved oxygen, heavy metals, and microbial contamination fall outside its scope, and laboratory testing remains necessary for bacteria and virus detection. Second, since labels are derived strictly from WHO pH and TDS thresholds, any contaminant that does not influence these two parameters will go undetected by the model. Third, pH probes are subject to drift and require regular recalibration against pH 4.00 and 6.86 buffer solutions, as neglecting this maintenance will degrade classification accuracy over time. Fourth, the current deployment relies on a single ESP32 node, meaning that monitoring multiple water points would require the introduction of a message broker such as MQTT to coordinate data collection. Fifth, the system has no local storage fallback, so any interruption in Wi-Fi connectivity results in data loss unless a microSD card or a retry queue mechanism is incorporated into the hardware setup.

7.3. Future Enhancement

Several concrete extensions are proposed to guide the future development of the system. Expanding the sensor suite to include turbidity, temperature, dissolved oxygen, and ORP would significantly broaden the range of detectable water quality problems beyond the current pH and TDS measurements. A Flutter-based mobile application with Bluetooth Low Energy support is proposed to deliver real-time alerts directly to users' phones, improving accessibility in field conditions. On the infrastructure side, migrating the backend to a cloud platform such as AWS or DigitalOcean would enable remote monitoring and improve overall system uptime and reliability. Forecasting capabilities are also envisioned through the integration of time-series models such as LSTM or Prophet, which would allow the system to anticipate water quality issues hours before they manifest. To improve transparency and user trust, explainable AI techniques such as SHAP or LIME are proposed so that each prediction can be accompanied by a breakdown of which sensor readings most influenced the outcome. Finally, a scheduled weekly retraining pipeline is recommended to ensure the model remains accurate and responsive to evolving water quality patterns over time.

7.4. Conclusion

This is a low-cost IoT and AI-powered water monitoring system designed for Nepal, integrating an ESP32 sensor node, a Node.js/MongoDB backend, a Random Forest model served via FastAPI, and a React dashboard. Tested on 4,451 samples, the system achieved 99.78% accuracy, with results validated as stable and free from data leakage, as labels are derived from rule-based logic. Safety is enforced through a dual-layer mechanism combining the Random Forest classifier with a WHO-rule override that prevents hazardous readings from ever being marked safe, and on 891 held-out samples, only 2 low-confidence errors were recorded. The system successfully meets its core objective of real-time, low-cost water quality classification into four WHO safety tiers, and is architecturally designed for future extension with additional sensors and cloud features.

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